



Service selection mechanisms in the Internet of Things (IoT): a systematic and comprehensive study

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Abstract

Internet of Things (IoT) describes embedded devices (things) with Internet-based connectivity, enabling them to receive and send data through the communication network. In addition, it provides infrastructure to let things have interacted with each other and people. This advantage of the IoT can increase reliability, sustainability, and efficiency by enhanced information access fashion. This technology can be used in various fields, such as environmental monitoring, home, and building automation and smart networks. Furthermore, the main aim of Service-Oriented Architecture (SOA) is to select the best services among a pool of services. These services can be selected statically or dynamically regarding the service functionalities and performance limitations. Since the performance of complex services is very important in many distributed domains. This study aims to systematically review the service selection mechanisms in the IoT. The service selection mechanisms are classified into centralized, decentralized, and hybrid classes. Also, the detailed evaluation of these techniques brings a good suggestion for further studies.

Keywords Internet of Things · Web service · Service · Selection · Centralized · Decentralized · Quality of service

1 Introduction

The main goal of the Internet is human–human interaction, and recently, it makes the thing–thing interaction as well [1–3]. This new communication technology is called Internet of Things (IoT). The IoT is a connection of physical objects that can be observed and controlled over the Internet [4, 5]. In the IoT, many things can be reached using many actuators and sensors [6, 7]. By implanting mobile things into gadgets and everyday objects, different forms of communication type between people and things has appeared and a new dimension to the information and communication world is added [8]. In this context, many possible application scenarios such as smart homes and offices, e-health, and learning activities can be realized [9, 10]. In addition, from the business users' viewpoint,

many values will be noticeable in many fields such as automation, logistics, inventory control, business process management, and intelligent transportation [10–12]. However, there are some challenges associated with the IoT, such as trust and security, data gathering, integration, standardization, network-related issue, service management, and governance requirement [13–15].

In fact, in the context of the IoT, there are different sources connected to the Internet providing many services such as cameras, sensors, actuators, mobiles, homes, hospital, etc. [16, 17]. Hence, it is responsible to provide the right service, in the right place and at the right time [18]. In IoT, the selection of services and features among many similar services is a complex process [19–21]. As the number of things or devices is increasing, the number of services is also increased and the selection process becomes more complicated [22–24]. To address the service selection challenge, the service provider needs some tools to characterize user preferences and control relevant services [25, 26]. The service selection method is, essentially, based on different criteria known as non-functional attributes. These attributes mainly known as Quality of Service (QoS) attributes [27]. There are three main challenges in

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the IoT service selection issue. First, the service selection method must be adaptable according to user preferences. Second, it must return the most appropriate service for the requestor. Third, in the context of the IoT, scalability is an important issue and must be considered in selecting the services [16].

Since the number of things and services on the IoT is increasing, selecting an appropriate service is very important. Therefore, studying and investigating the important mechanisms in this area is very important. While a Systematic Literature Review (SLR) is a detailed strategy to recognize and study all related works about a particular area, this paper carries out an SLR-based approach for reviewing the selection mechanism in the IoT. The SLR builds a correct evaluation on service selection mechanism using trusted and exact papers. These features help the researchers to have exact conclusions and make precise decisions [28, 29]. Therefore, reviewing and analyzing the current service selection techniques are the main aims of this paper. Briefly, the main aims of this paper are:

- Presenting an SLR and studying the current approaches for service selection in the IoT.
- Describing the main challenges about the IoT service selection.
- Dividing the service selection mechanisms into centralized, decentralized and hybrid mechanisms.
- Providing an exact evaluation of discussed mechanisms using some important QoS metrics.

Section 2 explains the related works. The primary concepts and a review process are described in Sects. 3 and 4, respectively. Section 5 discusses the selected service selection mechanisms in three main classes. The comparison among service selection mechanisms is provided in Sect. 6. Section 7 provides open issues. Finally, the conclusion is provided in Sect. 8.

2 Related work

Bouzary and Chen [30] have surveyed service selection and composition methods in cloud manufacturing. They have discussed the selection criteria, algorithms, optimization functions, relationship consideration, and mapping methods in detail. In addition, they have found that the investigated methods focused on some performance metrics such as selection criterion and cost, time, availability, and reliability. Moreover, some other service performance measures, such as throughput, service coverage, carbon emission, and pass-rate are utilized in some other studies. The main benefits and flaws of each cloud service selection technique are also highlighted. However, it is not related to the IoT domain. Also, a systematic study is missed.

Yang et al. [31] have surveyed and classified service selection mechanisms based on user preferences and deployment in a multi-cloud environment. They have described some components in multi-cloud environment architecture such as service-oriented application, deployment manager, and local scheduler. Also, some performance measures such as availability, reliability, throughput, and cost are considered for evaluating the discussed methods. The main features of each service selection and deployment techniques are also described. However, IoT is not the investigated environment of this study. In addition, papers selection process is not defined.

Moreover, Sobhika [32] has reviewed many approaches and architectures for service selection for ranking of the services. In addition, they have reviewed some QoS ranking estimation methods like neighborhood-based approach, ADF,¹ and local optimization for cloud services ranking using the service usage experiences of consumers. The major aim of the discussed mechanisms in this study is to find the individual services' rank. The main features of each service selection technique are also reviewed. However, the paper has some weakness. It is not about the IoT. The process for selecting the papers is not clear. Future work has not been provided. Finally, a tabular comparison among the article is missed.

Furthermore, Khédiri and Zaghdoud [33] have focused on the uncertainty requirements for cloud service discovery and composition. They have provided a set of methods in the literature for reviewing and categorizing based on risk modeling. Also, a set of features are recommended as risk modeling which is: type of service, type of uncertainty, response time, and robustness of the process. However, the IoT and service selection are not discussed in this paper. The way of selecting the papers and SLR are not provided. In addition, future work has not been provided.

Guerfel et al. [34] have investigated existing cloud services composition and discovery mechanisms. They also have discussed an architecture of cloud services composition based on SOA architecture. They have discussed the discovery approaches on two descriptions: syntactic descriptions and semantic descriptions. Also, they have explained the level of SOA layers within an organization, such as integrated composition level, business process, business services, IT services, and processing infrastructures. Finally, they have proposed a new architecture for service composition based on SOA architecture. This survey suffers from some issues. The papers selection rule is not discussed. Newly published papers have not been discussed. Also, it only focused on cloud computing and the

¹ ADF is the abbreviation of Adjusted cosine, Data smoothing and Fusion of similarity.

papers are not classified. Finally, open issues are not discussed.

Asghari et al. [35] have focused on service composition methods that were applied in the IoT environment. The aim of this study is to analytically and statistically categorize and analyze the present IoT service composition methods (published between 2012 and 2017). Five important QoS factors have been defined by authors which are availability, scalability, response time, cost, and reliability. Service composition methods in the selected studies are classified into two main categories with a focus on functional and non-functional properties aspect. The main features and advantages of each IoT service composition technique are also illustrated. However, the paper is not related to the service selection in the IoT. Finally, new papers have not been discussed.

Finally, Asghari and Navimipour [36] have discussed the existing optimal cloud composition methods by considering the running time. In this paper, existing methods have been presented in two categories: single and multi-cloud environment. Also, they have analyzed service composition mechanisms in each of these categories. Existing methods on the multi-cloud environment were compared on the dataset mentioned in previous papers. Also, recommendations for new working areas are suggested. The main features and advantages of each cloud service composition technique are also illustrated. However, the paper is not related to the IoT and service selection. The papers are not selected clearly. Finally, new papers have not been discussed.

3 Background

This section is about the related terminologies in the field of IoT and service selection.

3.1 IoT architecture

Internet has extremely changed in the last years by connecting many things. These things may have dissimilar sizes, abilities, computational power and can support any types of applications [11, 37]. Mainly, IoT has a 3-layer architecture including perception, network, and application layers, as depicted in Fig. 1 [38].

- Perception layer: It is like the five sense organs of IoT, which is generally recognizing objects and gathering the information [38, 39].
- Network layer: It is known as a brain of IoT which is responsible for transmitting and processing information.
- Application layer: It is a top layer of IoT architecture and links the users and applications [38, 39].

The applications of the IoT include every aspect of our daily life, such as natural disasters prediction, industrial applications, water lack monitoring, smart homes, medical applications, smart security, smart metering and monitoring, smart cities, intelligent transport system design, agriculture application and so on [10, 40, 41]. In contrast to all of these advantages, IoT has some limitations and challenges, such as naming and identity management, standardization and interoperability, privacy, protection and security, confidentiality, green IoT, service management, and network-based issues [42, 43].

3.2 Service selection

Service selection as one of the main parts of service management tries to select the geographically separated and deployed services. The service selection architecture is shown in Fig. 2. The users are separated into some categories corresponding to the geographically positioned services [44]. The service selection architecture has three parts: job submission, scheduler and execution monitor [45–47]. Job submission is responsible for accepting the job request. A scheduler is responsible for communicating with the UDDI² registry to search a suitable service provider [48]. Finally, the execution monitor must monitor the execution state of the request.

For comparison of service selection techniques in the IoT, some QoS criteria will be used, such as reliability, processing time, execution time, availability, stability, response time, flexibility, user preference, accuracy, scalability, performance, load balancing, and security issue. Therefore, a brief discussion of these metrics are as follow:

- Reliability: It is the possibility that a request is properly handled in the expected time or not [49, 50].
- Processing time: It is another important factor which depends on the requested tasks and ability of things [51, 52].
- Execution cost: It is a pay for using a service to get a result by requester [53].
- Execution time: It can be assessed by means of analytical modeling, case-based reasoning, and history data and shows the total time of execution [54].
- Availability: Service availability is the probability that a service is available or not [49, 55].
- Stability: It is the ability of the system to provide a normal response even in the case of some problems [56].
- Response time: It is a time from submitting a job until getting the results containing the communication and running time [16].

² UDDI (Universal Description, Discovery, and Integration) works as a mediator between the requester and the provider.

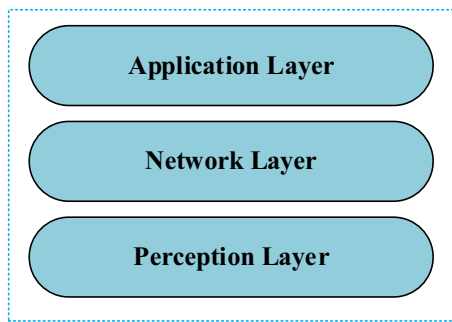


Fig. 1 The layered architecture of the IoT

- **Flexibility:** Meeting users' various requirements by available services and resources is known as flexibility [57].
- **User preference:** It is some of QoS metrics that the users want to receive the requested service based on those metrics. On the other word, these metrics have higher priority for users than other metrics [58].
- **Accuracy:** It is a metric for showing the accurate performance of service evaluation and selection methods [59].
- **Scalability:** It is a desirable property of a network to show its ability to handle growing amounts of requests [22].
- **Performance:** Performance is the practice of evaluating the metrics and benchmarks for IoT systems. It can determine how well an IoT system is functioning and what improvements can be made to the network [22].
- **Load balancing:** It tries to allocate tasks among multiple servers consistently [60–64].
- **Security:** Security issues refers to the policies, applications, and controls utilized to protect data, services, and the infrastructure of the IoT environment [53].

4 Research procedure

This section presents SLR about the service selection methods in the IoT. It is a serious evaluation of all research that addresses a particular issue [65]. In this section, it is utilized to achieve a comprehensive study of the service selection mechanisms in the IoT.

4.1 Question formalization

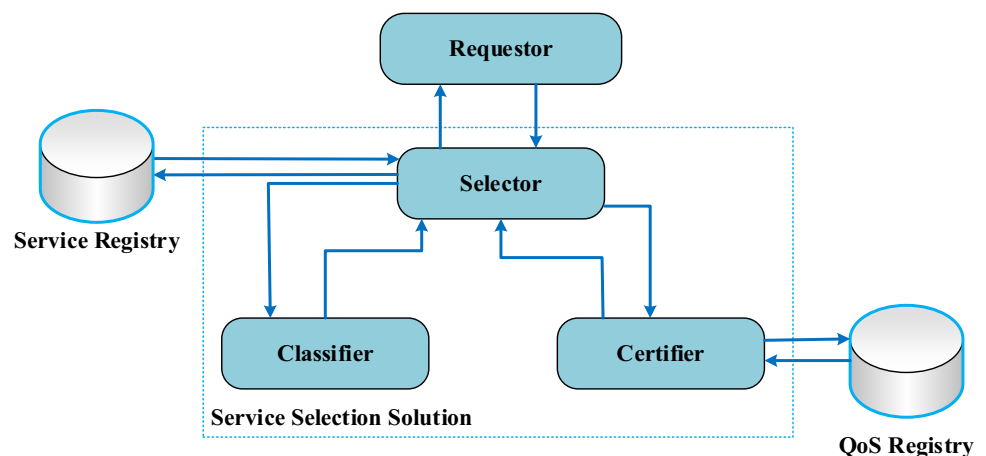
The present research goals are recognizing and analyzing all related papers that have examined in service selection in the IoT. Another aim of this paper is understanding the main issues and challenges in the field of IoT service selection, which includes reliability, processing time, execution time, availability, stability, response time, flexibility, user preference, accuracy, scalability, performance, load balancing, and security issue. This paper addresses the following Research Questions (RQs):

- RQ1: What is the position of the service selection component in the IoT?
- RQ2: What is the service selection itself?
- RQ3: What are the important QoS parameters in the service selection?
- RQ4: How is the searching process for assessing the IoT service selection methods?

4.2 Article selection process

The process of papers selection has three main steps [66–70]. In the first step, Google Scholar is employed as the main search engine to find related papers based on “allintitle” phrase and some keywords, including (service selection, IoT, internet of things, service select). Consequently, 27 papers are found from automatically search. The distribution of these papers by year of publication from

Fig. 2 Service selection architecture [48]



2013 to 2018 is shown in Fig. 3. Furthermore, Fig. 4 shows the classification of them in any publisher. Finally, Fig. 5 illustrates the distribution of the papers over time.

Next step starts by defining some criteria to assure that only worthy publications will be used. The student papers, reports, commentaries, appraisal studies, and non-English papers are excluded [71–75]. Eventually, 23 papers are considered for more analysis. Then the selected papers are reviewed to confirm their relevance. The subject, publication year, and journal rank are other attributes to include or exclude the papers. After applying filters, 17 papers are obtained, which are:

- certainly published in IoT field,
- explained the proposed method evidently and clearly,
- improved some of the considered parameters.

Finally, the classification of the selected service selection papers is done. The papers can be divided into centralized, decentralized, and hybrid methods. Figure 6 illustrates the classified papers and the description of each category. Five papers out of 17 papers are about the centralized mechanism (Table 1); 8 out of 17 papers are dealt with a decentralized mechanism (Table 2), and 4 out of 17 papers are dealing with a hybrid mechanism (Table 3). In addition, selected papers based on the publication year are shown in Figs. 7 and 8. The papers classification in each category is shown in Fig. 9.

5 Service selection mechanisms in IoT

The selected service selection mechanisms, their difference, advantages, disadvantages, evaluation parameters, and architectures are examined in this section. Also, in this section, the three selected classification including, centralized, decentralized, and hybrid for IoT service selection are described.

5.1 Centralized mechanism

This section describes the centralized mechanism and reviews the selected mechanisms. In Sect. 5.1.3 the mechanisms will be compared.

5.1.1 Overview of a centralized mechanism

On one hand, Information Systems (ISs) are the vital tools of many organizations in managing and coordinating commercial processes, and on the other hand, IoT is a paradigm can connect physical things to the Internet [53]. The IoT infrastructure can be integrated with ISs for removing the limitations of the physical world and the business systems [83]. With the growth of IoT, the number

of associated things and their offered services is growing quickly. Therefore, selecting a suitable service that satisfies a user's requirements turns into a time-consuming problem [84]. The centralized service selection mechanisms are one of the primary mechanism to perform the service selection process in the IoT. Generally, the centralized systems have a central component to collect the information about requirements of service requests and the providers' status. With this information, an optimal solution can be obtained [85]. The centralized systems might know significant information such as the capabilities of service providers [86].

5.1.2 Review of the selected centralized mechanisms

Na, et al. [76] have presented a method to balance energy on IoT devices by selecting appropriate services. For deploying IoT services, this method uses a platform-independent middleware based on the service-oriented paradigm. According to the proposed scheme, the middleware balances energy using intelligent service selection to evade overloading of some things. In the proposed method, to provide a dynamic service selection process, authors have presented a game-based model where each component can make a service selection decision independently. This model for IoT service selection consists of the following components: 1. The player which selects a service, 2. The population which denotes the set of players selecting services, and 3. The strategy which related to a certain selection decision. A player wants to increase the lifetime, which will extend the lifetime of the associated application. Generally, in the game setup, there is a centralized controller to collect payoffs of all players via their current strategies and to send the average payoff to all players. Then each player can make the service selection decision using its average and present payoff. The evaluation metrics are system size that affected the execution time, service diversities that affected the performance, service utilization rates that affected the execution time and cost. Based on performed evaluation and comparison, low execution time, and low cost are the main benefits of this method. But, it suffers from low flexibility, therefore this issue can affect the communication energy, especially in the group-based services selection.

Furthermore, Qi et al. [59] have proposed a "frequency–location–time"–aware IoT service selection based on historical records. In this paper, three factors on service selection process have been considered, namely, invocation time, location, and frequency, which affected the accuracy of IoT service selection. Therefore, a centralized service selection approach is needed to consider these factors simultaneously. The proposed method consists of three steps, first, each historical record of an IoT service is

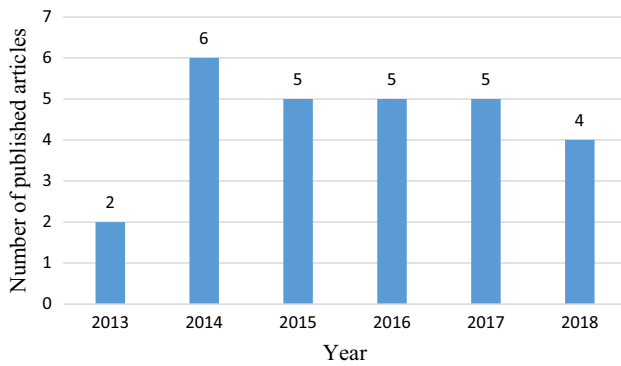


Fig. 3 Distribution of papers by years of publication

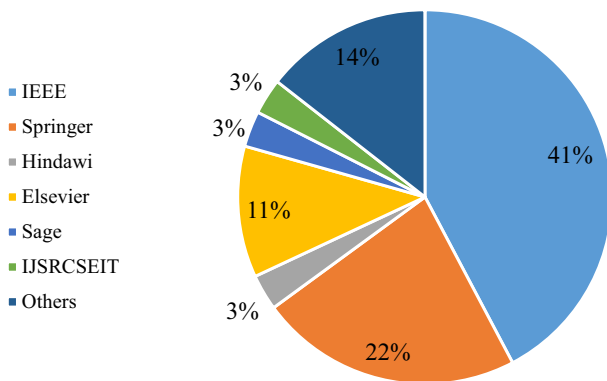


Fig. 4 Percentage of published papers based on different publishers

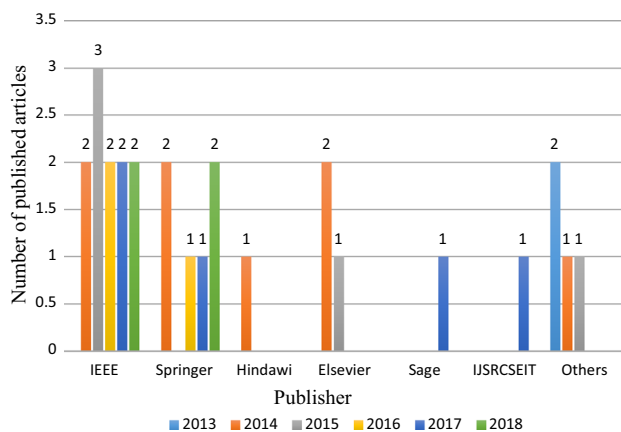


Fig. 5 Distribution of papers by the publisher

weighed using its service invocation time and location. Then, each candidate IoT service is weighed using its invocation frequency; lastly, it is evaluated and the quality-optimal one is delivered to the requester. Accuracy, execution time and execution cost are the evaluation metrics in this paper that have been analyzed by authors. According to the obtained results, high accuracy, low execution time, and low execution cost are the main advantages, but, the

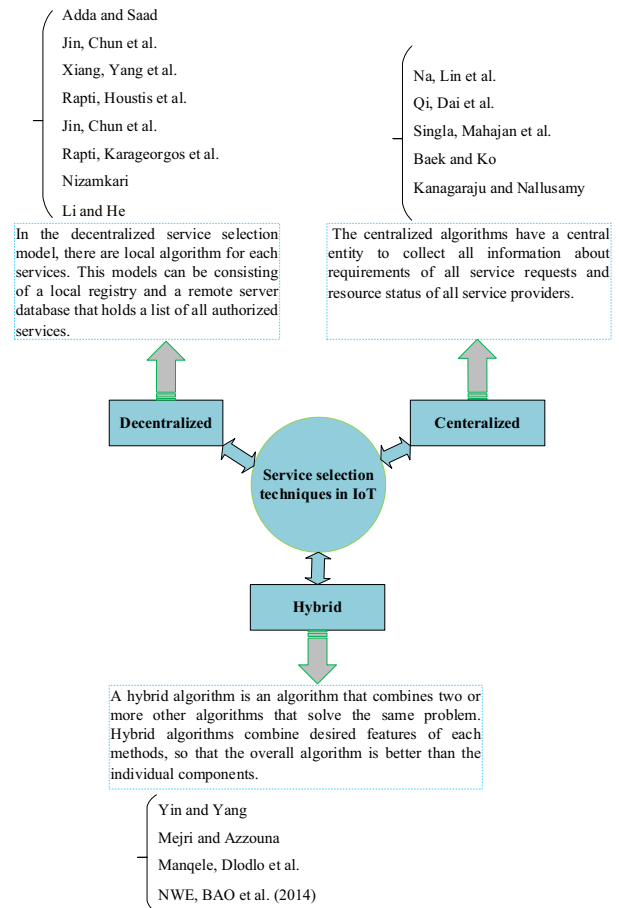


Fig. 6 Classification of the service selection techniques in IoT

proposed method suffers from some limitations, such as high computational power and network overheads.

Moreover, Singla et al. [19] have discussed the integration of the IoT and cloud computing to achieve efficient computation offloading process. Also, they have claimed that the integration of IoT with the cloud solves many current challenges by providing extra features such as ease-of-use, ease-of-access, and low deployment costs. Hence, they have presented a model based on the Fuzzy Analytic Hierarchy Process (FAHP) and Fuzzy TOPSIS (FTOPSIS) to choose an optimal service in the presence of uncertainty. The proposed method enables decision-makers to better understand the total evaluation process and thus provides a more accurate and efficient decision support mechanism. Good efficiency and high security are the main advantages of the proposed approach. But, it does not consider other QoS metrics, that is a disadvantage of this method.

Also, a spatially-cohesive service selection to deal with highly dynamic status in mobile IoT has been proposed by Baek and Ko [77]. They have proposed a reinforcement-learning agent to select services in a dynamic way while optimizing some metrics. The service hand-over is a

Table 1 Details of the selected centralized papers

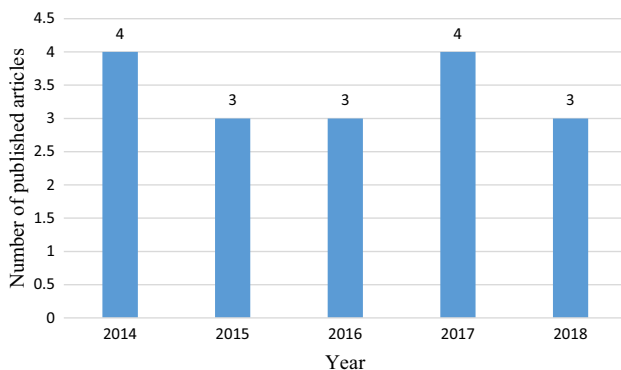
Publisher	Paper	Journal/Conference
IEEE	Na et al. [76]	e-Business Engineering
Sage	Qi et al. [59]	Distributed sensor networks
Springer	Singla et al. [19]	Lecture Notes on Data Engineering and Communications Technologies
Springer	Baek and Ko [77]	ICWE 2018
Springer	Kanagaraju and Nallusamy [18]	Cluster computing

Table 2 Details of the selected decentralized papers

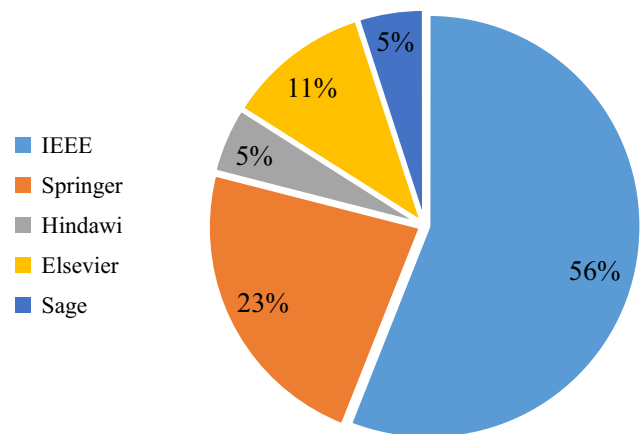
Publisher	Paper	Journal/Conference
Elsevier	Adda and Saad [57]	Procedia Computer Science
IEEE	Jin et al. [51]	International Conference on Service-Oriented Computing and Applications
IEEE	Xiang et al. [78]	Tsinghua Science And Technology
Springer	Rapti et al. [79]	Service Oriented Computing and Applications
Springer	Jin et al. [53]	Information System Frontier
IEEE	Rapti et al. [80]	International Conference on Services Computing
IEEE	Nizamkari [81]	International Conference on Inventive Systems and Control
IEEE	Li and He [82]	Special Section on Collaboration for Internet of Things

Table 3 Details of the selected hybrid papers

Publisher	Paper	Journal/Conference
Hindawi	Yin and Yang [56]	Journal of Sensors
IEEE	Mejri and Azzouna [16]	International Journal of Computer Applications
IEEE	Manqele et al. [22]	International Journal of Computer Applications
Elsevier	NWE et al. [58]	The Journal of China Universities of Posts and Telecommunications

**Fig. 7** Distribution of the selected papers by years of publication

concept of performing service selection dynamically and migrate service instances from device to device, to allow users to consume the services in a continuous manner even though the availability of services fluctuates. They have proposed a metric that measures how the selected services

**Fig. 8** Percentage of selected papers in each publisher

are located in a spatially cohesive manner in terms of the associating IoT devices. However, the algorithms were designed based on heuristics of greedy choices on the

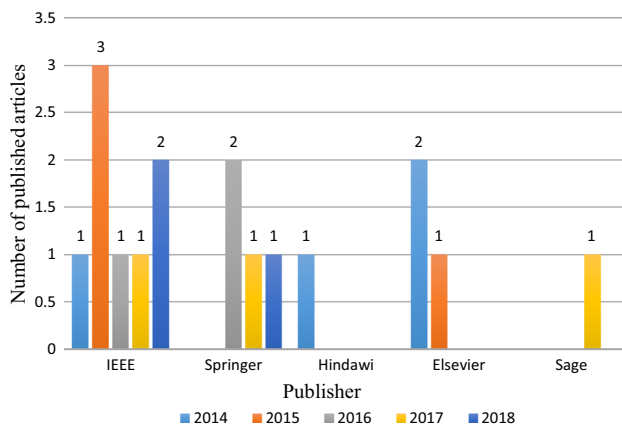


Fig. 9 A number of selected papers between 2013 and 2018

services that are associated with the nearest IoT devices, so makes decisions shortsighted. To deal with the limitations of the previous work, the authors have adopted reinforcement learning, which is a class of simulation-based machine-learning techniques. Reinforcement learning is about training a smart agent that learns the optimal policy of taking actions in a certain type of environments to achieve a given goal, by simulating a number of trials and errors. Main advantages of applying reinforcement learning are the maximization of long-term rewards, robustness, and model-free learning. First, the proposed service selection agent learns to optimize metrics such as the spatio-cohesiveness in a long-term manner rather than considering an only the current snapshot of the environment, which is challenging in highly-dynamic IoT environments. Second, the proposed agent is robust, which means that the service selection can be done even though the environment is observed partially, or encounters unexpected situations such as failures. Third, the proposed agent learns to optimize metrics without knowing the internal model of environments, which result in a great efficiency to deal with heterogeneous IoT environments and potential to extend. In this paper, the spatio-cohesive service selection problem has been formulated as a reinforcement learning problem by representing the selection process as a partial-observable Markov decision process. Then, they have used the actor-critic algorithm to solve the problem. The proposed method has capable of optimizing the spatio-cohesiveness and the number of hand-overs in a long-term period of executing user tasks. Scalability and stability are the main advantages of this method. This method does not consider some QoS metrics, such as security, availability, user preferences and etc.

Finally, Kanagaraju and Nallusamy [18] have studied an IoT architecture for industrial applications and have presented a Register Service Selection (RSS) method based on security architecture to collect the essential data and

transmit it to a cloud-based backend. In this work, RSS-based security algorithm is improved. Moreover, they have advised a generic security model for clouds to satisfy safety demands. The proposed architecture for industrial applications has three layers: frontend, gateway, and cloud back-end. The front-end layer is the sensing nodes that are included in four modules: RaspberryPi controller, the flow rate environmental sensors, interfacing circuits, and a wireless communication module. The front-end nodes can gather the sensor data and transmit it to a gateway. Then the gateway sends the gotten data to the cloud servers in the back-end. In other words, the gateway acts as a central controller in this method. Authors have used some metrics such as controlled value, settling time and rising times for evaluating the proposed method. High-security level, high availability, and good throughput are the main advantages of this method, and also this method has a good result for security issues. However, low accuracy is a limitation of this method.

5.1.3 Summary and comparison of the reviewed mechanisms

In this Section 5 selected centralized mechanisms about IoT service selection are analyzed. Originally, centralized models have one global registry where each service registers itself and then by one algorithm, process the registered services. The benefit of having one global registry is that the managing of resources is done centrally. The authority control is simply managed and the service description is consistency in style [87]. Authorization and security concerns are also handled easily as they are managed centrally [88]. However, this approach cannot scale well as the number of registered resources in the global registry is limited. Moreover, parallel accesses to the registry are limited and the response times may be high in a high load [89]. These methods have been able to solve the current service selection methods limitation by applying the QoS metrics. But, some parameters such as computational power, network overheads, and other QoS metrics are not been considered. In the above classification of service selection methods, all methods have centric algorithms or centralized model (such as UDDI register). These algorithms have a central authority to gather vital information of all service requests and the service providers. However, the centralized mechanisms have the scalability problem to manage a large number of requests. According to our survey, the low scalability is a similar limitation between all methods that may affect the other QoS metrics. Also, in the proposed centralized methods, security issues have less been considered. Low execution time, low execution cost, good accuracy, good stability, high availability are the main advantages of centralized methods. Table 4

Table 4 A comparison of the centralized mechanisms for service selection

Article	Techniques	Advantage	Disadvantage
Na et al. [76]	• Balance energy consumption model	• Low execution time • Low cost	• Without considering flexibility
Qi et al. [59]	• Time–location–frequency–aware method	• High accuracy • Low execution time • Low execution cost	• High computational power • High network overheads
Singla et al. [19]	• Fuzzy analytic hierarchy process	• Good efficiency • High security	• Without considering other QoS metrics
Baek and Ko [77]	• Spatially-cohesive service selection by adopting a reinforcement learning technique	• Good stability	• Without considering security issues, availability, and user preferences
Kanagaraju and Nallusamy [18]	• Register service selection based security architecture	• High security level • High availability • Good throughput	• Low accuracy

shows a comparison of the important features of each paper.

5.2 Decentralized mechanism

This section discusses the decentralized mechanisms for service selection in the IoT. Also, the selected mechanisms will be compared and examined in Sect. 5.2.3.

5.2.1 Overview of decentralized mechanism

The number of things in the IoT grows quickly, therefore, some challenges such as heterogeneity, scalability, and interoperability have raised. Consequently, service-orientation has emerged as a promising solution to tackle the complexity of IoT. Centralized methods may be good for a limited number of services, but they are not suitable for a large-scale environment. In the decentralized service selection model, there is a local algorithm for each service. This model can be having a local registry and a remote server database that holds a list of all authorized services [88]. On the other hand, in the decentralized models, the service provider can use chain models of service selection approaches. In this model, each local registry or algorithm

can directly be connected to the next register or algorithms [88]. The advantage of them is that there is no central registry management. Issues regarding security, complexity, and authorization are managed for each service independently. In the rest of this section, the solutions for solving the service selection issue via decentralized mechanisms are discussed.

5.2.2 Review of the selected decentralized mechanisms

A sharing approach and a Domain Specific Language (DSL) for IoT service discovery, consumption and selection have been proposed by Adda and Saad [57]. The model is based on a defined service-centric approach and has abstract and theoretical concepts. It does not use a central server/entity for service delivery and discovery. Also, they have proposed a set of tools to support the theoretical model in two parts: a DSL called IOTCollab, and an Eclipse Integrated Development Environment (IDE). Service has some information such as the type of data, how often these data are gathered, and the geo-location of the physical device, etc. In this model, an actor denotes a set of IoT objects which are managed by the same user. Therefore, an actor is signified by a set of IoT

objects and the related services. To select a service, the pre-selection process was needed. Then a global rating value is related to each service and the values of the path length and the average rating of each actor are considered. Authors in this method have proposed a query-response model that can manage the access control by a straightforward whitelist/blacklist policy. This can be considered as an advantage for this method. This method does not consider security issues such as identification, authentication, and data integrity.

Also, Jin et al. [51] have presented a Physical Service Model (PSM) to define heterogeneous IoT physical services. Device, resource, and service are the three core concepts of PSM. Based on PSM, four QoS attributes are defined and candidate services are rated based on user desires, which these attributes are processing time (PT), available time (AT), service area (SA), and RePutation (RP). On the other hand, for selecting the dynamically aggregate individual QoS ratings and selecting physical services, they have proposed a Physical Service Selection (PSS) method. Based on the PSM model, they have analyzed the mentioned four QoS attributes. Also, they have proposed an algorithm to aggregate the QoS ratings according to a user preference to select a physical service. PSS has pre-sorting, filtering, and final-sorting phases. In the pre-sorting phase, they have taken a user preference, such as [AT, SA, PT, RP], which means that AT is the most preferred QoS attribute and RP is the least one. In addition, sorting of individual QoS ratings may result in high time complexity. To solve this weakness, they have used a pre-sorting method. They have also assessed the performance of the proposed approach in terms of the execution time, meeting a user request and preference for physical services. The performance of the method is also evaluated in terms of available time, service area, processing time, and reputation. In addition, they have used spatio-temporal attributes for evaluating the proposed method. High availability, low processing time, optimal service area, and good reputation are the advantage of this method. However, this method does not consider user preferences, privacy, and security issues.

Moreover, Xiang et al. [78] have investigated a QoS-based service selection method. In service-oriented IoT, real-world devices provide their functionalities as web services that do not have a centralized section for managing the service selection process, therefore this issue can be affected the selection accuracy and time complexity. Hence, they have tried to solve the current methods of limitations by proposing the new method. In the proposed method, first, they have presented a lightweight description method called Three Particular Predicates (TPP) to describe QoS. Furthermore, based on the TPP, they have formalized the QoS-based service selection problem, and it

is decomposed into a sub-problem. Then a QoS-based service matching algorithm has been presented. Especially, checking the similarity of parameters and value of predicates are done for similarity prediction. The traces-driven simulations have shown that the method has increased the matching precision and the recall rate in comparison with current methods. According to the obtained results, scalability and low time cost are the power of the method. Also, it does not consider the other QoS metrics that can be affected the method performance, therefore the low flexibility is the disadvantage of this method.

Furthermore, Rapti et al. [79] have proposed a decentralized bio-inspired IoT service discovery and selection approach. The proposed method is inspired by the Response Threshold Model (RTM) and does the selection of atomic services. RTM is inspired by decentralized insect colonies. In this work, RTM has mapped to the service selection problem and taking benefit of the natural capability of colonies to work without any central control. With this decentralized technique, two main desirable features in any artificial system are obtained (flexibility and robustness). Therefore, for defining required entities, an agent-based approach has adopted. Each Service Provision Nodes (SPN) is characterized by a decision-making entity to transmit user requests towards its neighbors. In this model, SPNs act using the bio-inspired RTM, and a user request is met by agent actions. The scalability and routing performance have been assessed by evaluating the average number of hops needed for a user request. Obtained results have shown that the method has efficient scalability and routing performance. Good scalability and good routing performance in the routing are the advantages of this model. According to the obtained results, when the topology of the network changed, this method had less flexibility and low performance.

Again, Jin et al. [53] have presented a method for IoT service selection using a physical service model. They also have proposed a PSM as a conceptual model for physical services descriptions. It defines physical services in terms of spatiotemporal features that are important factors in an IoT domain. Also, they have defined spatio-temporal, positive, and negative as the main QoS attributes, which reflected the physical services' features. Moreover, they have proposed a PSS method for evaluating the QoS values of CPSs according to user preference. This algorithm has three phases: pre-sorting, filtering, and final sorting. Also, they have evaluated the model based on reputation, processing time, execution cost, and reliability. Also, low execution time, cost, and good reliability are the main advantages of this model. But, the user preferences, privacy and security issues are not investigated well.

Also, Rapti et al. [80] have proposed a decentralized IoT service selection method based on Artificial Potential Filed

(APF), which are formed upon each user request to provide the services. The power of each APF depends on the percentage of the requested services that can be provided by the respective SPN. The APFs are adapted to select a set of services which are requested by a user. Their approach for selecting the requested services and forwarding user requests among the nodes is adapted from the interaction between electrically charged particles. The average number of hops required to meet a user request and the number of placed requests in the queue are two metrics that authors have used them for evaluating the proposed method in the decentralized environment. Simulation results have indicated that the method maintains satisfactory performance and scalability as the number of SPNs in a network grows. Also, a suitable load balancing of the requested services was reported. Therefore, high load balancing, high scalability, and good performance are the benefits of this method, but, it has low flexibility when the topology of the network changed.

Moreover, Nizamkari [81] have proposed a recommendation algorithm to solve a service selection problem in the IoT using graph-based Collaborative Filtering (CF). In this algorithm, a node can forecast its ratings for the nodes that are offering the desired service and then it can select the best-rated service provider. The proposed method for sending and receiving a service request and service rate to/of its friends uses a breadth-first approach. Also, this method calculates two kinds of influences: correlational and social. This method has three main steps, in the first step, a scalable method is needed for calculating predictions, where a node calculates its own forecasts instead of using a central system. In the second step, an algorithm is required to measure the impact of one node on another by means of a combination of trust and similarity between them. In the third step, Libimseti dataset is used for evaluating the performance of the proposed method. Also, two types of trust between nodes are calculated, if nodes are directly connected, they used a direct social trust. If they are not connected, they calculated an indirect social trust value via trust propagation and aggregation. This method can scale to match the network growth. High scalability, good flexibility, and high accuracy are the main benefits of this model. However, it and many CF-based models have one common limitation, they can not predict a rating for a node if nobody has rated it yet.

Finally, Algorithms for KFSR-IIoT (AKFSR-IIoT) to choose a key feature sequence of resource services in Industrial IoT (IIoT) has been studied by Li and He [82]. Generally, in a collaborative task, IIoT resource services are divided. The rest steps in this paper for IIoS service selection are as follows: First, resource service modeling is presented to obtain all the Resource Service Chains (RSCs). Next, the Degree of Influence (DI) is defined.

After normalization, the average DI of any two attached features is evaluated; thus, the key feature sequences are determined. Based on the DIs, an algorithm is presented to measure the synthetic DI of an RSC. Lastly, the AKFSR-IIoT is verified and good scalability is reported. But, it does not consider other QoS metrics.

5.2.3 Summary and comparison of the reviewed mechanisms

Decentralized mechanisms just rely on connections among service providers or between service providers and end users to solve the service selection problem, without any central entity [90]. Using such decentralized self-management principles does not generate optimal or near-optimal solutions in service selection. In this strategy, each server makes local resource allocation decisions based on received requests from clients, and clients coordinate with servers in one or more iterations to obtain a service selection solution [90]. Eight selected decentralized papers were discussed in the previous section. Also, the advantages and disadvantages of the selected mechanisms were presented. Decentralized methods do not have scalability limitation. Also, this method does not have any single point of failure, therefore, this method has high availability and reliability. On the other hand, decentralized methods have good load balancing between the providers and users. High availability, low processing time, optimal service area, efficient scalability, good flexibility, and high accuracy are the main advantages of the decentralized methods. Table 5 shows the comparison of the most important features of each paper.

5.3 Hybrid mechanism

This section defines the hybrid mechanism for service selection. Then, it reviews the selected hybrid mechanisms. Finally, the discussed mechanisms will be analyzed.

5.3.1 Overview of hybrid mechanisms

A hybrid mechanism mixes two or more mechanisms, either choosing one or switching between them [91]. It combines the desired features of each, so, the constructed algorithm is better than the basic mechanisms. As discussed earlier, the centralized methods have good performance in the small-scale network and on the other hand, for the large-scale network, decentralized methods are a good choice. Several approaches in centralized and decentralized service selection methods have been studied, but these studies have not considered the existed differences. Therefore, providers should be considered the advantages of centralized and decentralized methods and

Table 5 A comparison of the decentralized mechanisms in service selection

Article	Techniques	Advantage	Disadvantage
Adda and Saad [57]	• Data sharing strategy and a DSL method	• Low response time	• Without considering the security issues
Jin et al. [51]	• Physicals service model	• High available • Low processing time • Optimal service area	• Do not consider user preferences • Less considered the privacy issues
Xiang et al. [78]	• Three particular predicates method	• Scalability • Low time cost	• Low flexibility • Without considering other QoS metrics
Rapti et al. [79]	• Bio-inspired method based on RTM	• Efficient scalability • High routing performance	• Without considering user preferences and other QoS metrics • Low flexibility
Jin et al. [53]	• Physicals service model	• High reliability • Low processing time • Low execution cost	• Without considering user preferences • Without considering security issues
Rapti et al. [80]	• Artificial potential fields method	• High satisfactory performance • Scalable • Efficient load balancing	• Low flexibility
Nizamkari [81]	• Graph-based CF recommendation algorithm	• High scalability • Good flexibility • high accuracy	• Low performance in nodes ranking
Li and He [82]	• Selecting key feature sequence of resource services	• Scalability	• Without considering other QoS metrics

then use these features to propose a new hybrid method [92]. On the other hand, in the IoT, similar functional services are evolving in different QoS because of the widespread deployment of distributed things. With regard to IoT applications that can be centralized and decentralized, the hybrid methods should be considered for increasing the performance of the service selection methods by employing the benefits of both of them.

5.3.2 Review of the selected hybrid mechanisms

Yin and Yang [56] have studied the Web service selection problem from the candidate set. They also have formulated the selection of Web services for smart things as single-source many-target shortest path problem on MapReduce. In addition, a pruning algorithm for Dijkstra and breadth-first search has been proposed in this paper by authors. Since the MapReduce programming model becomes more and more popular in data-intensive computing, they have evaluated some related algorithms on its open source implementation instance-Hadoop. However, when graphs become bigger and bigger, precise algorithms require much time to compute many nearest neighbors and some border nodes may waste more computing sources. Therefore, the suggested an approximation algorithm to handle this

problem. The empirical evaluation on real-life graphs and Hadoop platform have shown that the algorithm is more effective than the breadth-first search shortest path algorithm. The powers of the method are low execution time, low processing time, and good stability, and the limitation of it is low performance where the authors have suggested that using the approximation algorithms can handle this problem.

Also, NWE et al. [58] have proposed a Flexible QoS-based Service selection Algorithm (FQSA) to give subjective preferences for users in a friendly style. The FQSA selects the services via subjective information which is provided by objective information. It uses ANN back-propagation algorithm to discover the objective factors and uses similarity aggregation approach to assess the credibility of the user subjective factors. The proposed model has four main components, one service repository, and two service providers. Service repository manages the services selection process centrally. The workflow of the model is initiated when a user request is received. Firstly, service matching process matches the functional service explanations of the accessible services gained from UDDI repository. Then, the ranking process positions the candidate services based on their non-functional QoS values which are calculated by FQSA. Finally, the services which match

with the user expected QoS necessities are selected. Evaluation metrics used in this paper are QoS aggregation, QoS reasoning, QoS scalability, personalized confidentiality, and user-friendliness. The results have shown that scalability, personalized confidentiality, and user-friendliness are the main advantages of this method. However, this method does not consider user preference.

Manqele et al. [22] have proposed a preference-based IoT dynamic service selection for the smart campus. Based on this paper, the smart campus refers to an organization that utilizes provided services by IoT. IoT data must be saved intelligently and must be used for smart monitoring. According to the existing research, such as [93], authors have developed artificial intelligence algorithms that could be centralized or distributed based on the need to make sense of the collected data. Moreover, since recommender systems pre-select the interested services by a user, a recommender system is used in a smart campus environment. The proposed architecture consists of 4 steps. In the first step, the user preferences have been considered. The second phase of the algorithm includes matching the service and the user profile according to the attribute name-value pairs. In the third step, each service profile is matched against the user profile. Finally, in the fourth step, the result has notified to the user. The accuracy of the method has been evaluated based on recall, response time, and precision values. According to the experimental results, they have shown that the content-based algorithm acts better than CF. Also, the content-based algorithm can return relevant services and takes a smaller time as compared to the CF. The efficiency of this method is better than the CF-based methods. This method has a low response time and also considered user preferences. This paper has some limitations, such as, it does not consider the other QoS metrics that can be affected the service selection efficiency.

Mejri and Azzouna [16] have proposed a service selection method for IoT. They have used both artificial intelligence through the use of Artificial Neural Network (ANN) and multi-criteria analysis through the use of Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) model to provide the best service. Their approach is composed, mainly, of two phases: the prediction phase and the selection phase. First, in the prediction phase, an ANN predictive model is employed to estimate the QoS values of services. Thereby, it takes as inputs the contextual attributes of both users and services, those are extracted from the user request and from the description of the service in the registry. Moreover, since there are several different QoS attributes, several instances of the ANN run simultaneously to forecast the values of each QoS. Second, in the selection phase, they have adopted a Multi-Criteria Decision Making (MCDMs) method with parallel implementation, that relies on the QoS predicted values. It allows

the service requestors to express their preferences regarding QoS attributes. In this paper, for achieving the service selection based on user feedback, they have used the reputation mechanism. The main goal of this service selection method is using previous feedbacks of users which means a set of historical interactions to identify the best services or service providers among several ones. In order to control reputation, all members must cooperate, coordinate and communicate with each other. Moreover, they used a central QoS registry that has the responsibility of gathering and recording QoS information from web service consumers. Response time, reliability, execution time and scalability used as evaluation metrics. According to the obtained result, the proposed method was able to support all QoS features' types, generic as well as domain-specific attributes. In summary, low response time, high reliability, low execution time, and scalability are the main advantages of the method. However, this method suffers from low performance in the case of high searching space.

5.3.3 Summary and comparison of the reviewed mechanisms

The major advantage of hybrid methods for service selection is the increased efficiency. This method works very well even when the services are scarce. The hybrid methods structure is more flexible than centralized and decentralized methods. This helps in addressing all limitation easily, thus increasing the effective service selection between service providers and users. Hybrid service selection methods are formed considering the advantage and disadvantages of previous methods. Thus, existing methods with different features are mingled together which gives an opportunity to select optimal service for service requesters. This also results in minimization of service selection costs, as the benefit of methods and models can be shared. The major disadvantage of hybrid methods is the chance of having conflicts between some services or service providers. Since the hybrid method is scalable, therefore, the management of this method is difficult and it cannot consider the user preferences in some cases. 4 selected hybrid papers were discussed in the previous section. Also, the advantages and disadvantages of the mechanisms were presented. Low execution time, low processing time, good stability, high scalability, low response time are the main advantages for hybrid methods. In a general summary, it can be said that hybrid methods have a higher performance than centralized and decentralized methods and also can meet the requirements for more QoS. However, they may suffer from high complexity. Table 6 shows the comparison of the benefits and weaknesses of each paper.

Table 6 A comparison of the hybrid mechanisms in service selection

Article	Techniques	Advantage	Disadvantage
Yin and Yang [56]	• Shortest paths based web service selection	• Low execution time • Low processing time • Good stability	• Without considering scalability • Do not consider user preferences
NWE et al. [58]	• Flexible service selection algorithm using an artificial neural network	• High scalability • Personalized confidentiality • User friendliness	• Without considering user preferences
Manqele et al. [22]	• Preference-based IoT dynamic service selection	• Low response time	• Without considering other QoS metrics
Mejri and Azzouna [16]	• Self-Adaptive service selection method	• Low response time • High reliability • Low execution time • Scalable	• Less accuracy in large searching space

6 Discussion

Based on the performed SLR of service selection mechanisms, 56% of the papers among 17 selected papers have been published by IEEE. So, IEEE has the highest number of investigated papers. The selected papers are categorized into three categories that 5 of the 17 papers are centralized, 8 of them are decentralized, and 4 of the 17 are hybrid techniques.

In centralized systems, if the core controller (or algorithm) is damaged or is not available, all subsystem fail. The central algorithm has to be fast to make decisions between the service provider and service requester. Centralized methods do not scalable, because these methods do not have enough processing capacity. As the server can respond only a finite number of the service request, bottlenecks can appear when the traffic spikes. Also, these methods have good security than other methods. Since the centralized mechanisms have a centric managing section, therefore, execution time, response time, execution cost of them are low. Also, these methods have high security, high accuracy, and high availability. But, the centralized method has some. The abrupt failure of the entire system may occur. Less possibility of data backup and difficult server maintenance are other limitation of this mechanism. Finally, there is only one server node and because of availability reasons, it is inefficient and unprofessional to take the server down for maintenance.

In decentralized systems, every method makes its own decision. The final behavior of the system is the aggregation of the decisions of the individual methods. A decentralized method is based on a cooperation of subsystems, not controlled by the one sever. Advantages of decentralized system are scaling of system is made very simply that

you can add new services without interrupting of network or using extra bandwidth, (2) failing of a (or many) server(s) would not result in the failure of the system, (3) environmental adaptability is good, (4) less memory is used, (5) better options for exception handling, and (6) less capability for individual agents. However, the decentralized methods do not have the centric manager that manages requests and responses, therefore, it is difficult to know which node failed, so each node must be pinged for availability checking. Also, it is difficult to know which node responded and when a request is served by a decentralized system. Finally, it is actually hard to find out which node really served the request.

Hybrid mechanisms combine canalized and decimalized mechanisms to solve their problem. Hybrid mechanisms play an outstanding role in enhancing the search and selection capability of IoT. It combines the advantages of each canalized and decimalized mechanisms to establish a hybrid mechanism while trying to minimize any substantial disadvantage. Although hybrid mechanisms provide many advantages, some drawbacks do exist. (1) The hybridization process regularly creates extra components in the architecture, therefore, the complexity is increased. (2) In many works, hybrid algorithms seem to decrees the overall speed and in some cased the accuracy.

According to the surveyed papers, centralized mechanisms do not consider reliability, processing time, load balancing, response time, and user preference. But, in most papers, the security issues have been considered. However, the decentralized and hybrid mechanisms, unlike the centralized mechanisms, have considered reliability, processing time, load balancing, response time, and user preference. Security issues are not important in the decentralized and hybrid mechanisms, therefore, these

mechanisms do not consider security issues comprehensively. Moreover, in the hybrid mechanisms, the accuracy parameter does not have considered. All papers considered the scalability parameter in the proposed mechanisms. Service availability is the most important issue in the service selection mechanisms, but the authors do not attend this parameter well.

In the existed methods, the researchers compared and evaluated the metrics that have an effect on the service selection to find which factor is more significant in each one. Table 7 provides a summary of the discussed service selection methods and their main properties like reliability, processing time, execution cost, execution time, privacy and security issues, scalability, load balancing, routing performance, response time, and user preferences.

7 Open issue and future works

It is clear that there is not any mechanism, which solves all challenges related to service selection in IoT. For example, some techniques consider reliability, scalability, load balancing, routing performance, response time; but some considered processing time, execution cost, execution time, user preferences. Furthermore, some of the discussed mechanisms have used simulation to evaluate the proposed methods, while some other mechanisms have been evaluated in real-world scenarios. Also, some service selection techniques do not consider privacy and security issues. In addition, providing enhanced security to prevent the attacks and using the suitable protocol in the cyber-security context, which includes dangers, threats, and vulnerabilities management are very motivating difficulties. Furthermore, applying some new meta-heuristic algorithms such as shark smell optimization [94, 95] and particle swarm optimization [96–98] based on fuzzy logic [99–103] and neural network [104–107] for solving the service selection method in the IoT is very interesting line for future work.

In some techniques, user preference does not consider, therefore, the presentation of user preference-based methods can be presented as future work. While load balancing is one of the QoS metrics that can affect the performance of the IoT network, but some existing methods do not consider it. Also, in some techniques, user preference does not consider, therefore, the presentation of user preference-based methods can be presented as future work. Load balancing is one of the QoS metrics that affected the performance of the IoT network, that existing methods do not consider it. On the other hand, availability is one the critical feature where future research can focus more on it. Finally, since energy is one of the most important challenges of these days [108, 109], selecting the services with the aim of decreasing energy consumption is another good

line for future research. The comprehensive analysis of the most important challenges have been discussed as follow:

- **Response time:** communication between IoT smart devices can effect on the response time of service discovery and selection procedures to lessen the delivery time for final selected services. Things convergence and smart objects negotiations are main factors that can effect on the response time of final services procedure in the communication among IoT devices.
- **Cost:** service providers have an interactive business with their own costumers or consumers to achieve cost-effective process [110, 111]. One of the most significant challenges of this matter is that IoT smart devices can have competitive behavior for selling smart services. With respect to today's business competition atmosphere and financial considerations, the cost and price factor is a crucial concern for users to select them as well as the service providers to provide them [112, 113]. Therefore, providing cost-effective service selection methods which meet other required QoS factors should be considered in the studies.
- **Energy:** the association between battery life and coverage of objects can be improved by the distribution of load balancing. The energy consumption problem particularly becomes a significant matter when data transferring into a datacenter located anywhere far from the user physical location [114–116]. Discovering and selecting appropriate services from the nearest geographical cloud datacenters can minimize the level of energy consumption in the IoT environment. In addition, this challenge conducts to paying attention to service selection to use less number of IoT resources.
- **Trust, security, and privacy:** trust, security, and privacy management is a critical challenge for data access and data storage on any platforms [117–121]. The trust-based similarity methods provide a safe interconnection between smart objects and user requests to support political conclusions and permanency. The key challenge of this topic is that smart objects should support a safe condition for the critical environment such as health-care systems. In addition, the privacy issue is another main concern in IoT environment, which emphasizes on the protection of both users' and service providers' privacy and security precedencies. According to the pervasive use of numerous diverse services, it can be imaginable to deliver purposely or unintentionally some critical and personal users' or even providers' information along with requested services to the others who are not authorized to access them. The main challenge of this matter is that a protection model for users or service providers' privacy preferences should

Table 7 An overview of the discussed service selection mechanisms and their main features

	Centralised mechanism					Decentralised mechanism					Hybrid mechanism						
	Na et al. [76]	Qi et al. [59]	Singla et al. [19]	Baek and Ko [77]	Kanagaraju and Nallusamy [18]	Adda and Saad [57]	Jin et al. [51]	Xiang et al. [78]	Rapti et al. [79]	Jin et al. [53]	Rapti et al. [80]	Nizamkari [81]	Li and He [82]	Yin and Yang [56]	Mejri and Azzouna [16]	Manqele et al. [22]	NWE et al. [58]
Reliability	×	×	×	×	×	×	×	×	×	✓	×	×	×	×	✓	×	×
Processing time	×	×	×	×	×	✓	✓	×	×	×	×	×	×	✓	×	×	×
Execution cost	✓	✓	×	×	×	×	×	✓	×	✓	×	×	×	✓	×	×	×
Flexibility	×	×	✓	×	×	×	×	×	×	×	×	✓	×	×	×	×	×
Execution time	✓	✓	×	×	×	✓	×	×	✓	✓	×	×	×	×	✓	×	×
Stability	×	×		✓	×	×	×	×	×	×	×	×	×	✓	×	×	×
Security	×	×	✓	×	✓	×	×	×	×	×	×	×	✓	×	×	×	×
Scalability	×	×	×	✓	×	×	×	✓	✓	×	✓	✓	×	×	✓	×	✓
Load balancing	×	×	×	×	×	×	×	×	×	×	✓	×	×	×	×	×	×
Accuracy	×	✓	×	×	×	×	✓	×	×	×	×	✓	×	×	×	×	×
Availability	×	×	×	×	✓	×	×	×	×	×	×	×	×	✓	×	×	×
Response time	×	×	×	×	×	✓	×	×	✓	×	×	×	×	×	✓	✓	×
User preference	×	×	×	×	×	×	✓	×	×	×	×	×	×	×	×	✓	✓

be offered for contribution for protecting the critical information in the service selection process.

8 Conclusion and limitation

According to the SLR in service selection mechanisms from 2014 to 2018, the number of published papers is determined very high in 2017 and 2018. IEEE with 56% has the highest published papers. Elsevier and Springer with 11%, Sage and Hindawi with 5%, have the minimum number of papers. We classified 17 selected papers in three classes that five of them are a centralized method, eight of them are a decentralized method, and 4 of them are a hybrid method. Also, all selected approaches are compared using some factors such as accuracy, response time, reliability, execution time, scalability, reliability, processing time, execution cost, energy, latency, security, and load balancing. The results have shown that most of the papers try to improve response time, reliability, execution time, scalability, reliability, but user preferences, security and privacy issue are not considered in many papers. The results have shown that most of the papers tried to improve response time, reliability, execution time, scalability, and reliability, but, user preferences, security and privacy issue are not considered in many papers.

This paper has several limitations. Firstly, this paper only reviewed the paper which was extracted based on a keyword of “service selection in IoT, service selection in the internet of things”. Secondly, this paper is limited to search only five online databases. There might be other journals or publishers which may provide a more complete picture of the service selection in the IoT. Lastly, non-English journals were omitted from this examination. We believe that research regarding service selection in IoT techniques has also been discussed and published in other languages.

The decentralized mechanisms in comparison to centralized and hybrid mechanisms have high scalability. The decentralized and hybrid mechanisms have been able to reduce response time, and pay more attention to user preference issues. The hybrid technique has less attention from the researchers. Good user preference and low response time are considered as one of the main advantages of this mechanism, but these mechanisms do not consider load balancing. The centralized mechanisms are appropriate for designing secure service selection mechanism. Because of this mechanism more attention to security and privacy issues. Finally, the centralized mechanisms have lower reliability, lower user preference and high processing time compared to the other techniques.

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